

Neuropsychological Assessment Report

Subject: subj_001

Date of Report: April 13, 2026

1 Demographics and Clinical Context

Age: 38 years
Gender: M
Group: IBS
IBS Subtype: D
IBS Severity (SSS): 195

Patient belongs to the IBS group (n=65 of 105 total). IBS subtype: D (n=24 in cohort). Age 38 (cohort M=37.0, SD=11.8).

2 Domain Findings

2.1 Sleep — Bergen Insomnia Scale (BIS)

BIS total = 16.0 (Moderate). Cohort percentile: 59th. $z = +0.16$. This indicates clinically significant insomnia symptoms that likely impact daytime functioning and cognitive performance.

BIS Total: 16.0 Classification: Moderate
Cohort percentile: 59.3th Cohort z-score: 0.16

2.2 Sustained Attention — Continuous Performance Test (CPT)

CPT Detectability (d') T=60.0 (96th percentile in cohort, $z=+1.63$). Elevated metrics: Detectability mildly elevated (T=60). This pattern suggests difficulties with sustained attention and/or response control.

CPT Metric	T-score	Cohort %ile	Status
Signal detection sensitivity (d')	T=60.0	95.6	Elevated
Missed targets (inattention)	T=56.0	97.1	Normal
False alarms (impulsivity)	T=56.0	73.5	Normal
Mean reaction time	T=46.0	46.6	Normal
Reaction time variability	T=52.0	87.3	Normal
Anticipatory responses	T=45.0	11.8	Normal
Response speed consistency	T=47.0	62.4	Normal

Table 1: CPT-3 performance metrics. T-scores are normed ($M=50$, $SD=10$). Higher T-scores indicate less favorable performance across the displayed CPT metrics. Scores ≥ 60 are mildly elevated; ≥ 65 are clinically significant.

2.3 Fatigue — Chalder Fatigue Scale

Chalder Fatigue bimodal total = 3.0 (3/11 items endorsed). Classification: No significant fatigue. Cohort percentile: 45th. No significant fatigue indicated.

Bimodal total: 3.0/11 Classification: No significant fatigue
Cohort percentile: 45.2th Caseness threshold: ≥ 4

2.4 Emotional Distress — Hospital Anxiety and Depression Scale (HADS)

HADS Anxiety = 2.0 (Normal, 17th %ile). HADS Depression = 2.0 (Normal, 39th %ile). Emotional functioning within normal limits.

	Score	Classification	Cohort %ile	z-score
HADS-A:	2.0	Normal	16.7th	−1.06
HADS-D:	2.0	Normal	39.2th	−0.55

Cutoffs: 0–7 Normal; 8–10 Borderline; 11–21 Clinical caseness.

2.5 Neurocognition — RBANS

RBANS Sum Raw = 233.0 (57.4th cohort percentile). Interpretation is based on the cleaned dataset’s raw subtests and cohort-relative summaries rather than the older RBANS index-score fields. The raw RBANS profile is broadly mid-range across domains.

Index	Score	Cohort %ile	Classification
Immediate Memory	−0.27	34.8	Low cohort position
Visuospatial	0.56	73.0	Relative strength
Language	−0.06	41.7	Mid cohort position
Attention	0.42	71.6	Relative strength
Delayed Memory	0.53	73.0	Relative strength
Recognition	−0.27	28.4	Low cohort position
Total Scale	233.0	57.4	Mid cohort position

Table 2: *RBANS summary values in the standalone March 16 pipeline. Domain scores reflect cohort-standardized raw-subtest composites, and “Total Scale” reflects RBANS Sum Raw. Classifications describe relative position within the observed cohort rather than external age-corrected norms.*

3 Clinical Flags

Sleep (BIS):

- Clinically elevated insomnia
- Multiple items ≥ 3 days/week: Q3, Q4, Q5, Q6

Attention (CPT):

- Detectability mildly elevated (T=60)

4 Cross-Domain Pattern Analysis

Sleep-attention link: Insomnia co-occurs with CPT attention deficits. Sleep disruption degrades vigilance and sustained attention via reduced thalamo-cortical arousal regulation (Krause et al., 2017). Treating the sleep disturbance may improve attentional performance. Gut-brain axis involvement: As an IBS patient with elevated scores in sleep, this profile is consistent with gut-brain axis mediated effects. IBS-associated visceral hypersensitivity activates the hypothalamic-pituitary-adrenal (HPA) axis and alters serotonergic signaling, which can drive insomnia, fatigue, and affective dysregulation (Mayer et al., 2015). An integrated treatment approach addressing both gastrointestinal and neuropsychological symptoms is warranted.

5 Clinical Reasoning Chain

The following reasoning chain traces the step-by-step clinical logic used to integrate findings across domains, identify mechanistic links, and arrive at the recommendations below.

Step 1: Demographic and clinical context. M, age 38, IBS group, IBS subtype D. IBS-SSS=195.

Step 2: Sleep (BIS). Total score 16.0, classified as Moderate (cohort 59.3th percentile).

Step 3: Fatigue (Chalder). Bimodal total 3.0, No significant fatigue (cohort 45.2th percentile).

Step 4: Emotional distress (HADS). Anxiety 2.0 (Normal), Depression 2.0 (Normal).

Step 5: Sustained attention (CPT). Detectability T=60.0, Omissions T=56.0, Commissions T=56.0, HRT T=46.0.

Step 6: Neurocognition (RBANS). Sum Raw 233.0 at the 57.4th cohort percentile. The interpretation uses raw subtests plus cohort-standardized summary composites. Relative strengths are present in Visuospatial, Attention, Delayed Memory.

Step 7: Cross-domain integration. Flags span sleep, attention, supporting an interacting symptom pattern.

Step 8: Prognostic formulation. The prognosis depends mainly on whether the flagged domains respond to targeted follow-up.

6 Recommendations Based on Pre-Screening Results

Each recommendation below is explicitly linked to the pre-screening domain finding(s) that justify it. Domain labels in **bold** indicate the driving finding. Recommendations follow a two-stage structure: (1) determine whether further examination is necessary, then (2) outline treatment planning and interventions.

6.1 Further Examinations or Referrals Recommended

- **ATTENTION:** Repeat CPT or broader attentional testing if problems persist after sleep, mood, or fatigue interventions.

6.2 Suggested Therapeutic Interventions

- **SLEEP:** Prioritize CBT-I and structured sleep-hygiene measures, as insomnia is a likely upstream driver of daytime fatigue and reduced cognitive efficiency.
- **ATTENTION:** Add compensatory supports for sustained attention, such as task chunking, fewer distractions, and written reminders.
- **IBS INTEGRATION:** Coordinate neuropsychological follow-up with gastrointestinal care, since gut-brain symptoms may reinforce sleep, fatigue, and mood burden.

7 Visualizations

Three pairs of complementary figures are provided, each tailored to a specific audience. All scores are shown in the context of the full cohort (N=105), split by IBS status (n=65) vs. healthy controls (n=40).

7.1 For Clinical Peers

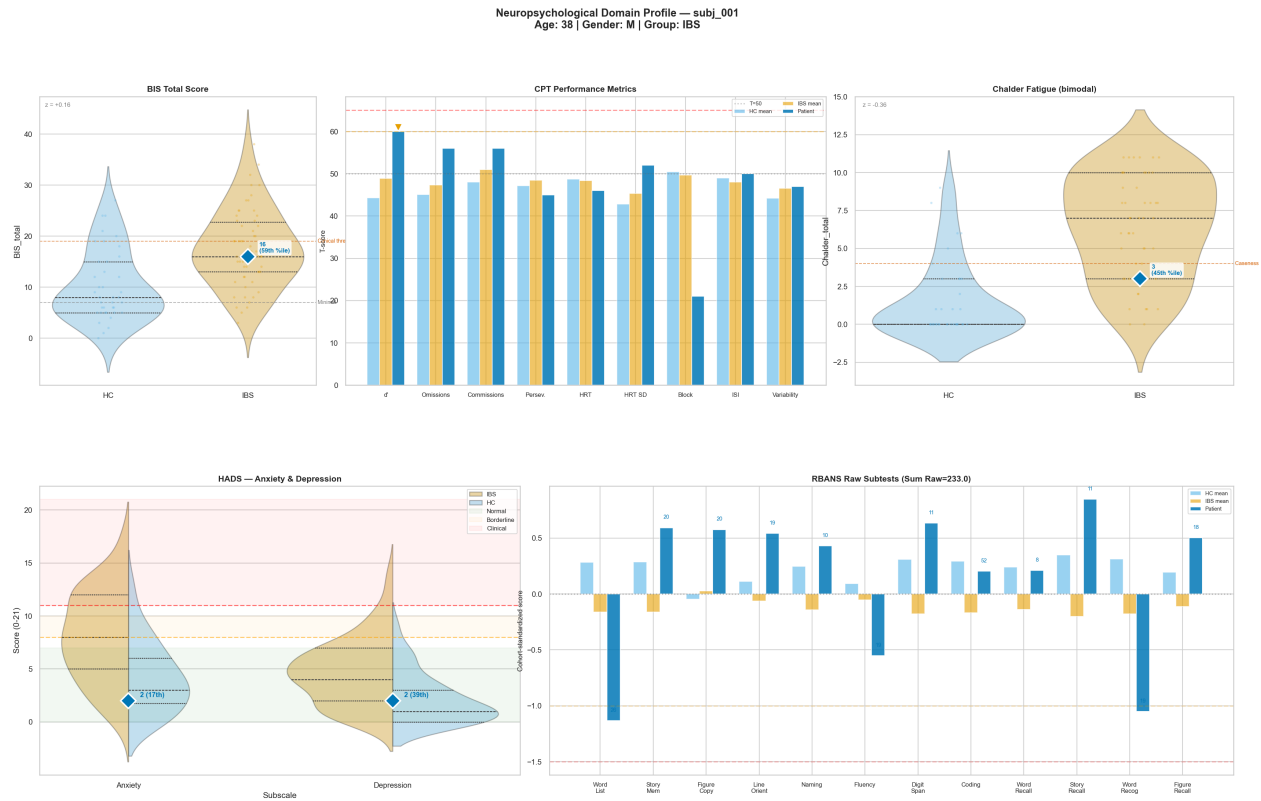


Figure 1: Multi-panel domain profile (clinical detail). Each panel shows the cohort distribution (violin plots split by IBS/HC group) with the patient's score marked as a blue diamond. **Panel 1 (BIS):** Insomnia total score with clinical threshold (dashed red, ≥ 19) and minimal cutoff (dashed gray). **Panel 2 (CPT):** Grouped bars comparing HC mean, IBS mean, and patient T-scores; red/orange triangles flag elevated metrics. **Panel 3 (Chalder):** Fatigue bimodal total with caseness threshold (dashed red, ≥ 4). **Panel 4 (HADS):** Split violin for Anxiety and Depression; shaded zones mark Normal (green), Borderline (amber), and Clinical (red) ranges. **Panel 5 (RBANS):** Raw RBANS subtests displayed on a common cohort-standardized scale, with the patient's raw score annotated above each bar and the broader RBANS Sum Raw shown in the title. Percentile ranks and z-scores annotate each patient marker. All distributions are within-cohort.

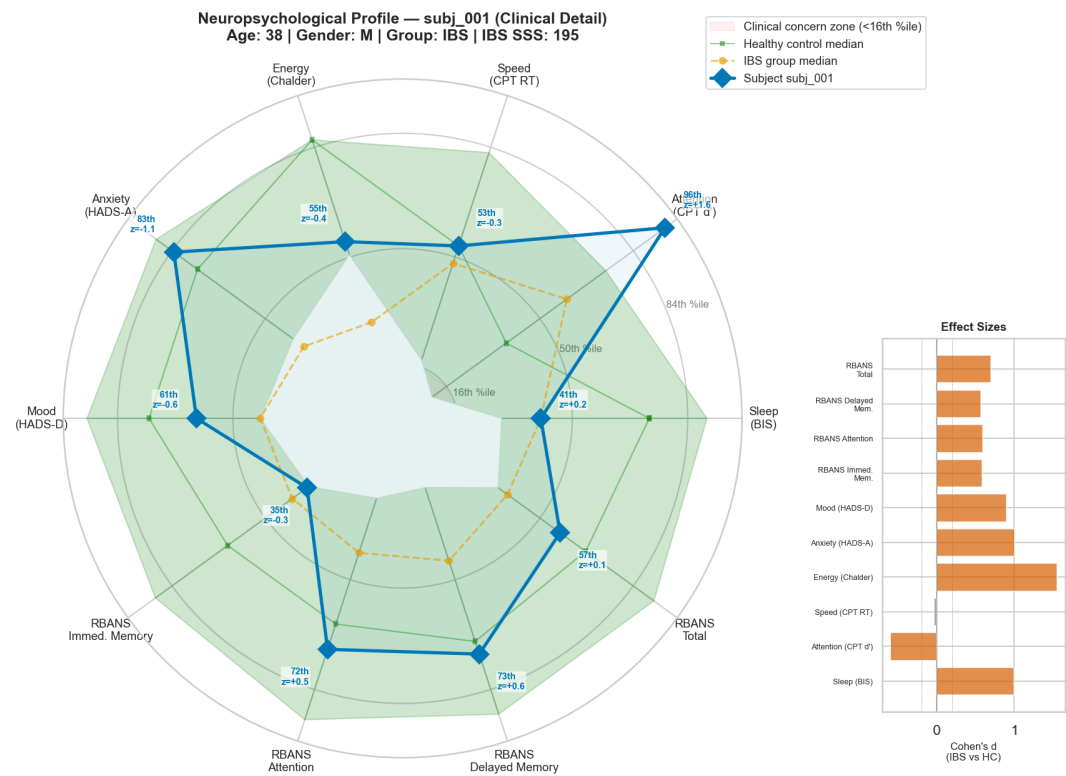


Figure 2: Radar plot (clinical detail, expanded 10-spoke version). Each radial axis represents one domain or subscale, transformed to a common 0–100 percentile-of-wellness scale (higher = better functioning; scales where high raw scores indicate worse outcomes are inverted). **Green band:** Healthy control range (16th–84th percentile, i.e. ± 1 SD). **Orange dashed line:** IBS group median profile. **Red center zone:** Clinical concern (<16th percentile). **Blue polygon:** Patient profile, with percentile and z-score annotations at each vertex. **Inset (right):** Cohen's d effect sizes for IBS vs. HC on each domain. Positive values indicate the IBS group performs worse. $|d| > 0.2$ (small effect) shown in amber; $|d| > 0.5$ (medium effect) in red. This inset contextualizes whether any patient deviations from the healthy range are consistent with the broader IBS group pattern or represent individual-specific findings.

7.2 For Referring Physicians

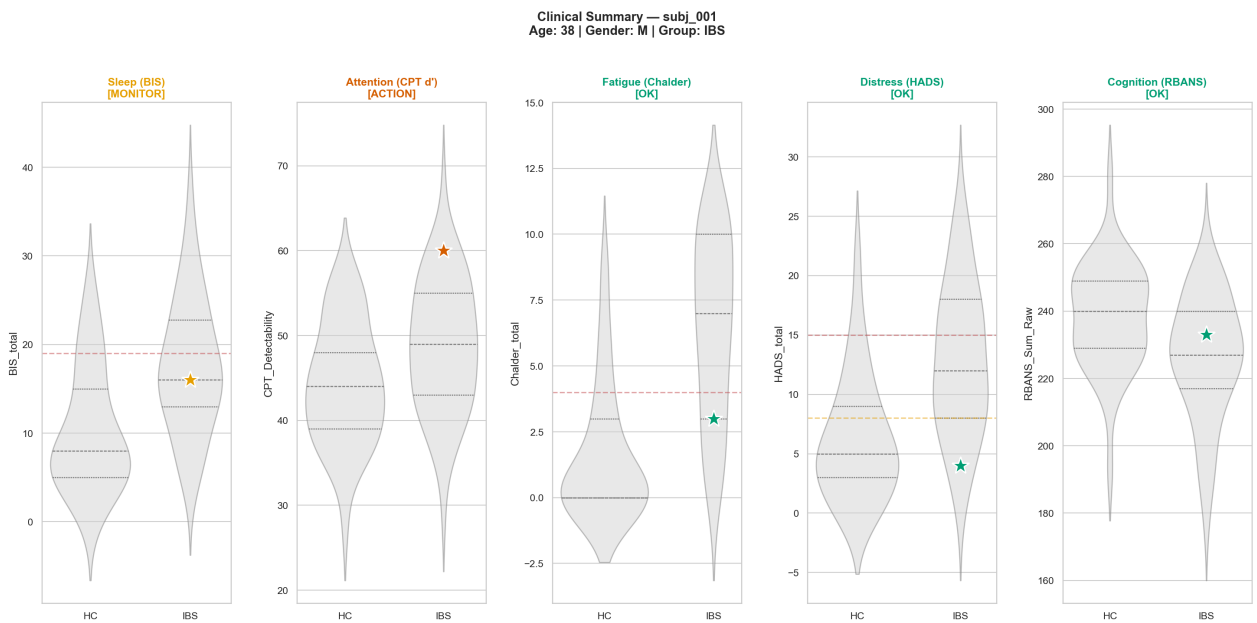


Figure 3: Clinical summary (physician view). Five-domain overview using simplified violin plots with traffic-light status labels: **OK** (≥ 50 th percentile of wellness), **MONITOR** (16th–49th percentile), **ACTION** (< 16 th percentile). The patient’s score is shown as a large star marker. This view is designed for rapid clinical triage: domains marked **ACTION** warrant immediate intervention or referral; **MONITOR** domains should be tracked at follow-up.

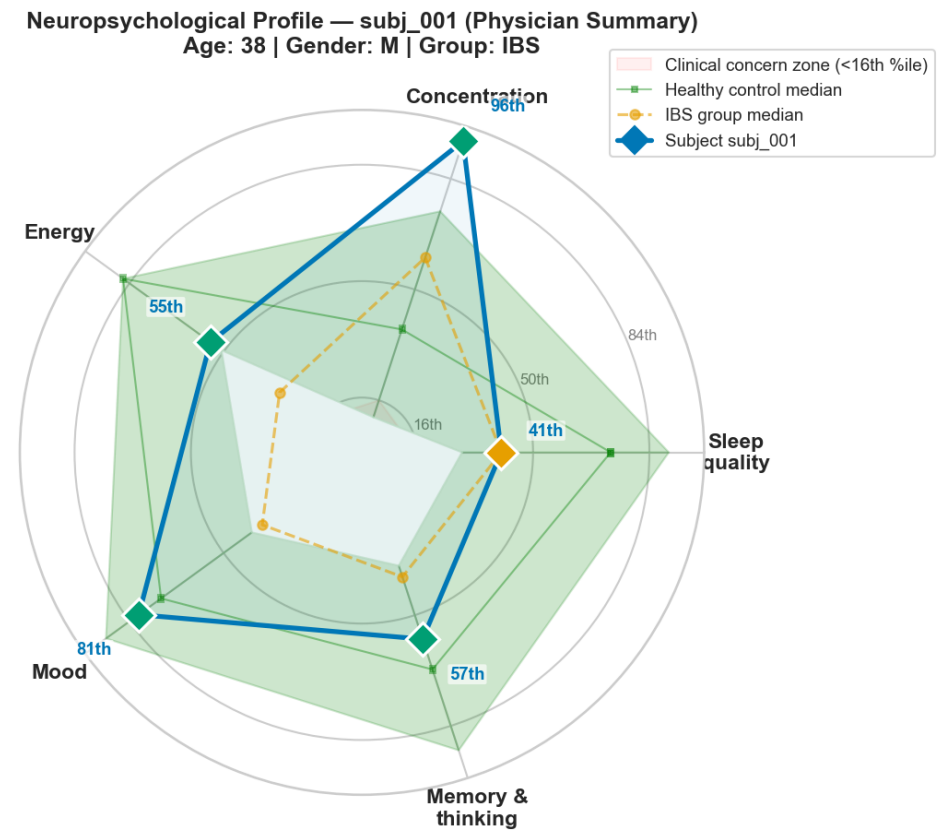


Figure 4: Radar plot (physician view, compact 5-spoke). Same percentile-of-wellness normalization as the clinical version, but using five composite domain spokes for rapid pattern recognition. Vertex colors follow the traffic-light system: green ($\geq 50\text{th}$), amber (16th–49th), red ($<16\text{th}$). The overall polygon shape reveals whether difficulties are focal (one spoke depressed) or diffuse (multiple spokes depressed). The green shaded band shows the typical healthy control range for reference.

7.3 For the Patient and Close Persons

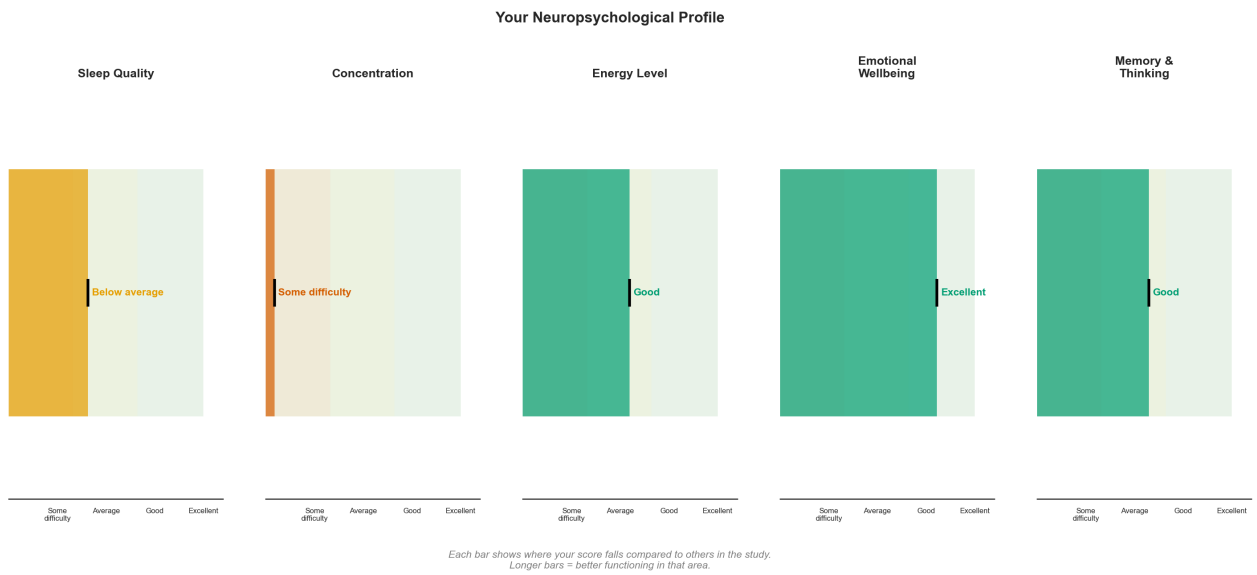
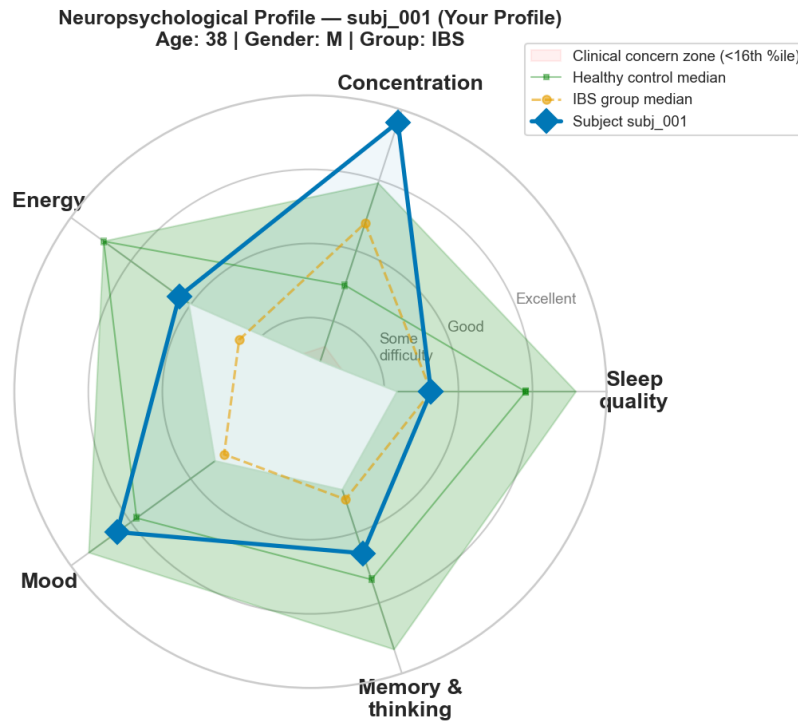


Figure 5: Your results at a glance. Each horizontal bar shows how your score compares to others in the study. Longer bars (reaching further right) mean better functioning in that area. The labels “Some difficulty”, “Average”, “Good”, and “Excellent” describe where your score falls. Areas shown in green are going well; areas in amber or red may benefit from support or follow-up. This chart covers five areas of health: sleep quality, concentration ability, energy levels, emotional wellbeing, and memory and thinking skills.



Your profile shows relative strengths in Concentration, Mood. some areas to monitor: Sleep quality.

Figure 6: Your overall profile. This spider chart shows all five areas together so you can see the overall shape of your strengths and any areas of difficulty. Points further from the center are better. The green shaded area shows the “typical healthy range.” The orange dashed line shows the average for people with IBS. The blue shape is your personal profile. The sentence at the bottom provides a plain-language summary of what the chart shows.

This report was generated by an AI-based neuropsychological analysis pipeline. All interpretations should be reviewed and validated by a qualified clinical neuropsychologist. Findings are based on cross-sectional cohort data ($N=105$). CPT values remain norm-referenced T-scores, whereas the RBANS section in this standalone March 16 pipeline uses raw subtests and cohort-standardized summaries rather than external age-corrected RBANS index norms. Clinical cutoffs follow established guidelines: HADS (Zigmond & Snaith, 1983; Bjelland et al., 2002), BIS (Pallesen et al., 2008), Chalder Fatigue Scale (Chalder et al., 1993), RBANS (Randolph, 1998), CPT-3 (Conners, 2014).